

NAME: KANISHK C

REG NO:192511278

EXP NO: 7

DEMONSTRATE VIRTUALIZATION BY INSTALLING TYPE-2 HYPERVISOR

AIM

To demonstrate virtualization by installing a Type-2 hypervisor on a host device, creating and configuring a virtual machine with a guest operating system using VMware Workstation.

PROCEDURE

Step 1: Download and install VMware Workstation on the Windows host operating system.

Step 2: Download a Tiny Core Linux ISO image to use as the guest operating system.

Step 3: Open VMware Workstation and select Create a New Virtual Machine.

Step 4: Select the downloaded ISO image file and configure the virtual machine.

Step 5: Set the required hardware configuration such as:

- Memory: 2 GB
- Processor: 1 CPU
- Hard Disk: 20 GB
- Network: NAT

Step 6: Give the virtual machine a name, such as Tiny Core Linux.

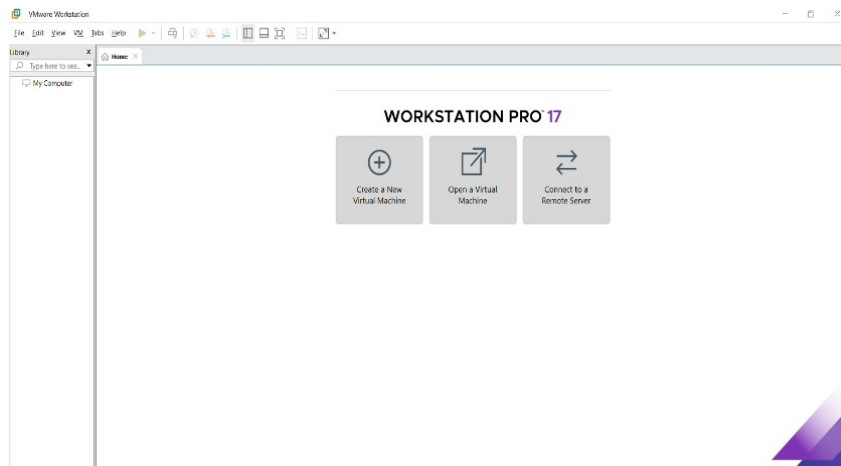
Step 7: Click Finish to create the virtual machine.

Step 8: Power on the virtual machine and allow it to boot from the Tiny Core Linux ISO.

Step 9: Select Boot TinyCore from the boot menu.

Step 10: Verify that the Tiny Core Linux desktop is successfully displayed.

OUTPUT



New Virtual Machine Wizard

Select a Guest Operating System

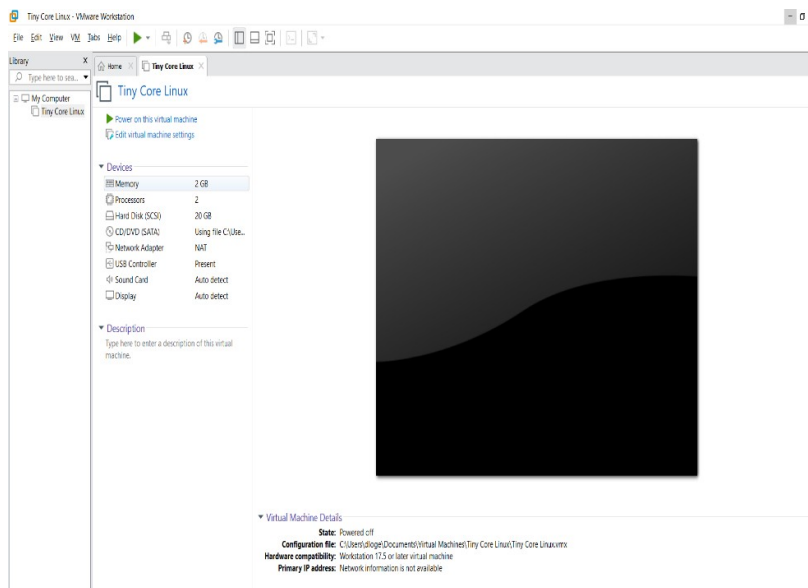
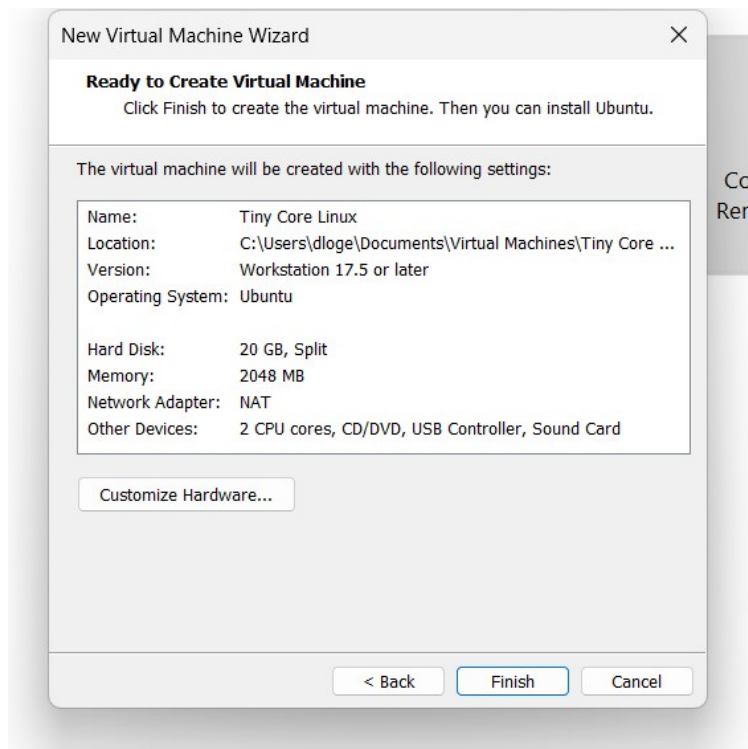
Which operating system will be installed on this virtual machine?

Guest operating system

- ☐ Microsoft Windows
☒ Linux
☐ VMware ESX
☐ Other

Version

- Ubuntu
Rocky Linux 64-bit
Sun Java Desktop System
SUSE Linux Enterprise 15 64-bit
SUSE Linux Enterprise 12 64-bit
SUSE Linux Enterprise 11 64-bit
SUSE Linux Enterprise 11
SUSE Linux Enterprise 10 64-bit
SUSE Linux Enterprise 10
SUSE Linux Enterprise 7/8/9 64-bit
SUSE Linux Enterprise 7/8/9
SUSE Linux 64-bit
SUSE Linux
Turbolinux 64-bit
Turbolinux
Ubuntu 64-bit
Ubuntu
VMware Photon OS 64-bit
Other Linux 6.x kernel 64-bit
Other Linux 6.x kernel
Other Linux 5.x kernel 64-bit
Other Linux 5.x kernel
Other Linux 4.x kernel 64-bit
Other Linux 4.x kernel
Other Linux 3.x kernel 64-bit
Other Linux 3.x kernel
Other Linux 2.6.x kernel 64-bit
Other Linux 2.6.x kernel
Other Linux 2.4.x kernel 64-bit
Other Linux 2.4.x kernel
Other Linux 2.2.x kernel



RESULT

The Tiny Core Linux virtual machine was successfully created, configured, and executed using VMware Workstation Type-2 hypervisor on the Windows host system.

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EXP NO: 8

CREATE A VIRTUAL MACHINE WITH 1 CPU, 2 GB RAM AND 15 GB STORAGE DISK USING A TYPE-2 VIRTUALIZATION SOFTWARE

AIM:

To create a virtual machine with 1 CPU, 2 GB RAM, and 15 GB storage disk using a Type-2 virtualization software (VMware Workstation).

PROCEDURE:

STEP 1: Download and install VMware Workstation on the host computer as a Type-2 hypervisor.

STEP 2: Download an Ubuntu or Tiny Core Linux ISO image for installing the guest operating system.

STEP 3: Open VMware Workstation and select Create a New Virtual Machine.

STEP 4: Select the downloaded ISO image file and choose the appropriate operating system.

STEP 5: Configure the virtual machine hardware with:

- Processor: 1 CPU
- Memory: 2 GB RAM
- Hard Disk: 15 GB
- Network: NAT

STEP 6: Give a suitable name to the virtual machine and select the location where the VM files should be stored.

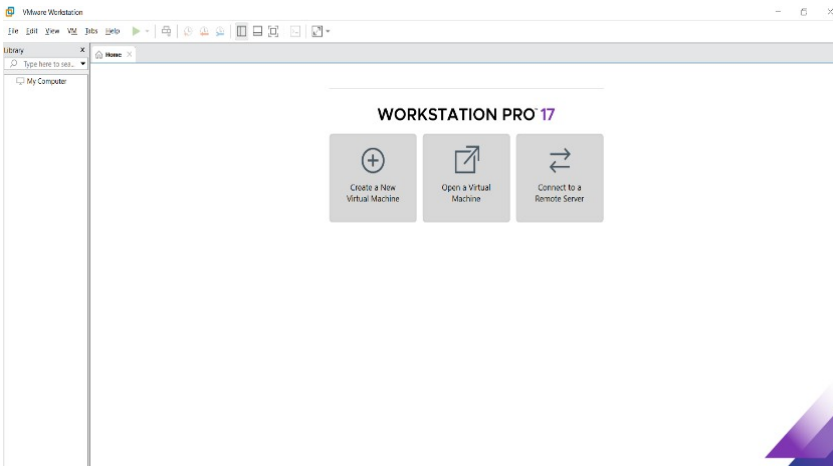
STEP 7: Click Finish to create the virtual machine.

STEP 8: Power on the virtual machine and allow the operating system to boot from the ISO image.

STEP 9: Complete the basic installation/configuration of the guest operating system.

STEP 10: Launch the virtual machine and verify that it is running successfully with 1 CPU, 2 GB RAM, and 15 GB storage.

OUTPUT:



New Virtual Machine Wizard

Select a Guest Operating System

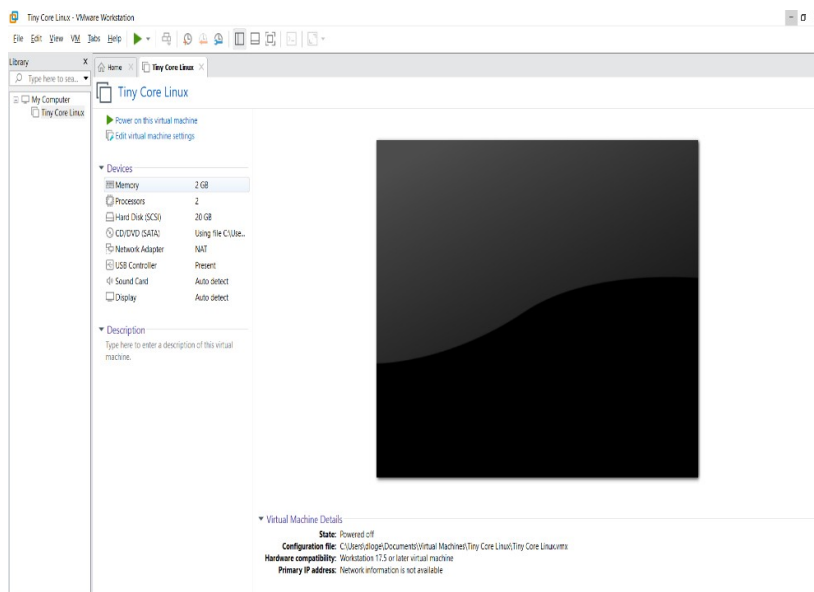
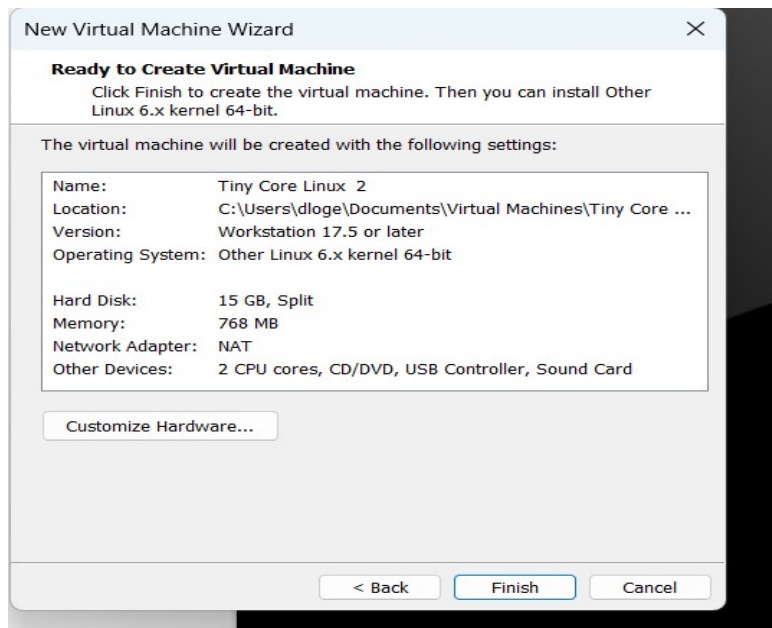
Which operating system will be installed on this virtual machine?

Guest operating system

- ☐ Microsoft Windows
☒ Linux
☐ VMware ESX
☐ Other

Version

- Ubuntu
- Rocky Linux 64-bit
- Sun Java Desktop System
- SUSE Linux Enterprise 15 64-bit
- SUSE Linux Enterprise 12 64-bit
- SUSE Linux Enterprise 11 64-bit
- SUSE Linux Enterprise 11
- SUSE Linux Enterprise 10 64-bit
- SUSE Linux Enterprise 10
- SUSE Linux Enterprise 7/8/9 64-bit
- SUSE Linux Enterprise 7/8/9
- SUSE Linux 64-bit
- SUSE Linux
- Turbolinux 64-bit
- Turbolinux
- Ubuntu 64-bit
- Ubuntu
- VMware Photon OS 64-bit
- Other Linux 6.x kernel 64-bit
- Other Linux 6.x kernel
- Other Linux 5.x kernel 64-bit
- Other Linux 5.x kernel
- Other Linux 4.x kernel 64-bit
- Other Linux 4.x kernel
- Other Linux 3.x kernel 64-bit
- Other Linux 3.x kernel
- Other Linux 2.6.x kernel 64-bit
- Other Linux 2.6.x kernel
- Other Linux 2.4.x kernel 64-bit
- Other Linux 2.4.x kernel
- Other Linux 2.2.x kernel



RESULT:

Thus, a virtual machine with 1 CPU, 2 GB RAM, and 15 GB storage disk was successfully created and executed using VMware Workstation Type-2 virtualization software.

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EXP NO: 9

**CREATE A VIRTUAL HARD DISK AND ALLOCATE THE
STORAGE
USING VMWARE WORKSTATION**

AIM:

To create a virtual hard disk and allocate the required storage using VMware Workstation.

PROCEDURE:

STEP 1: Open VMware Workstation and select the existing virtual machine.

STEP 2: Make sure the virtual machine is powered off.

STEP 3: Right-click the virtual machine and select Settings.

STEP 4: In the Virtual Machine Settings window, click Add.

STEP 5: Select Hard Disk from the Hardware Type list and click Next.

STEP 6: Select SCSI as the virtual disk type and click Next.

STEP 7: Select Create a new virtual disk and click Next.

STEP 8: Set the disk size to 2 GB as specified in the lab manual.

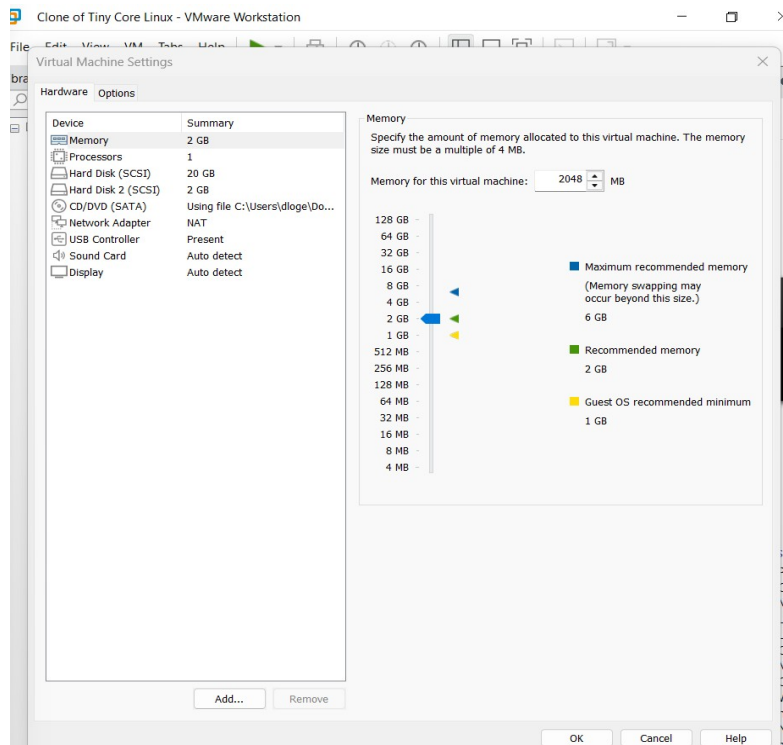
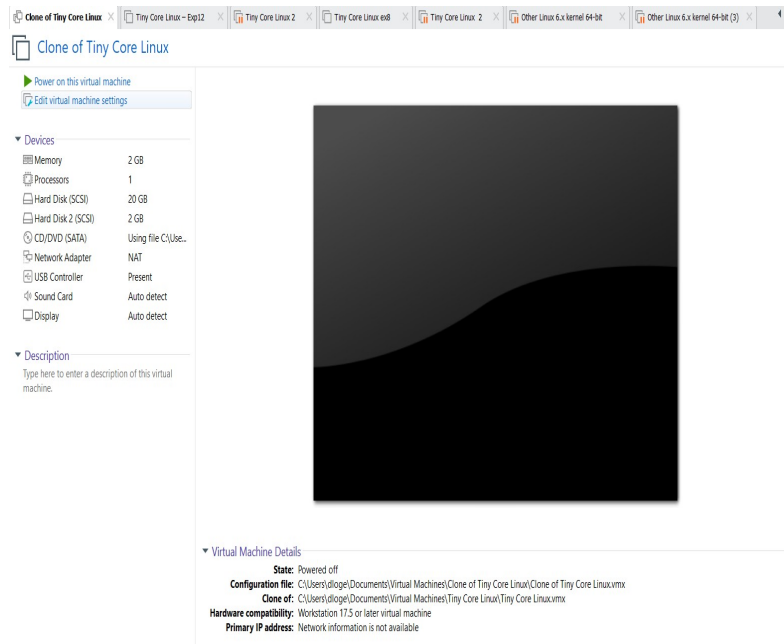
STEP 9: Select Split virtual disk into multiple files and click Next.

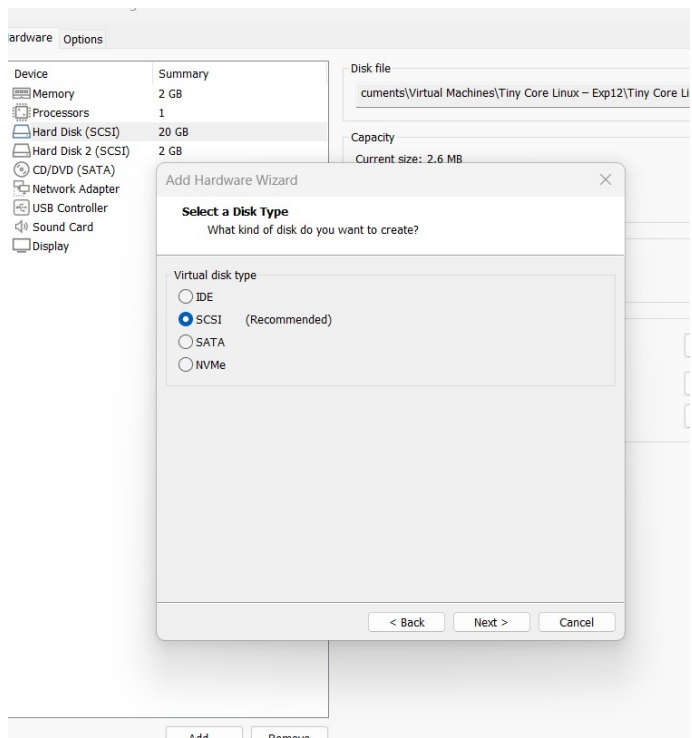
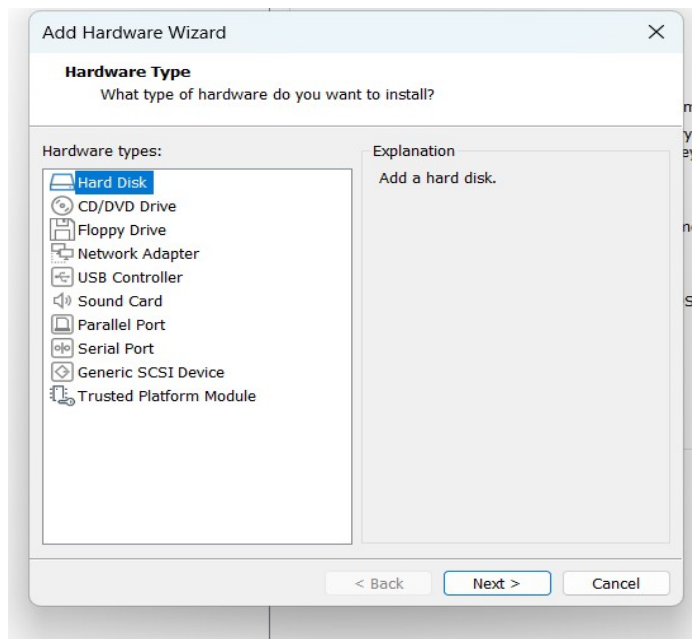
STEP 10: Give a suitable name for the virtual disk and click Finish.

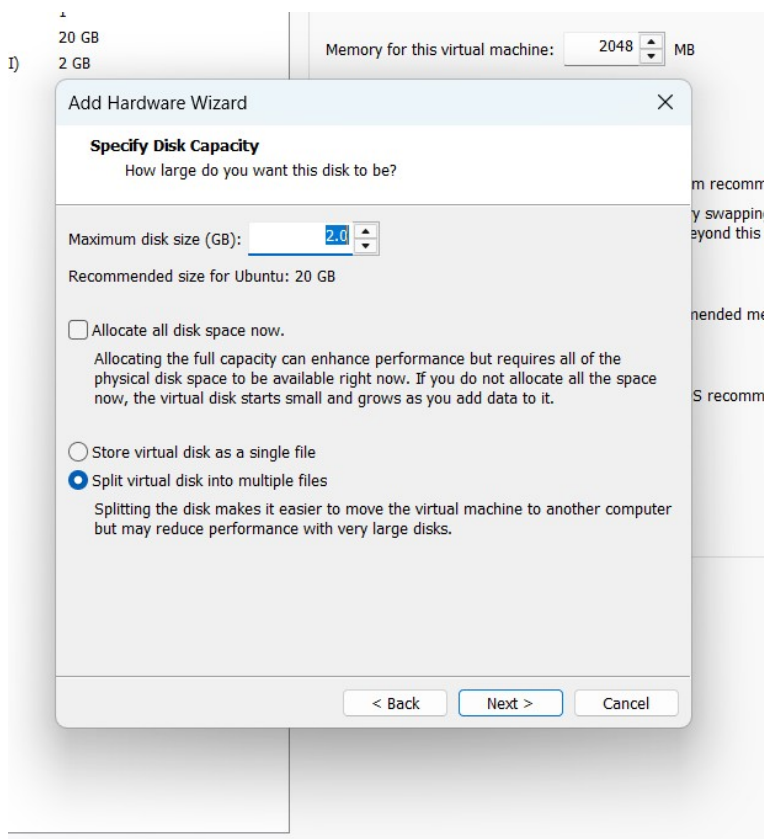
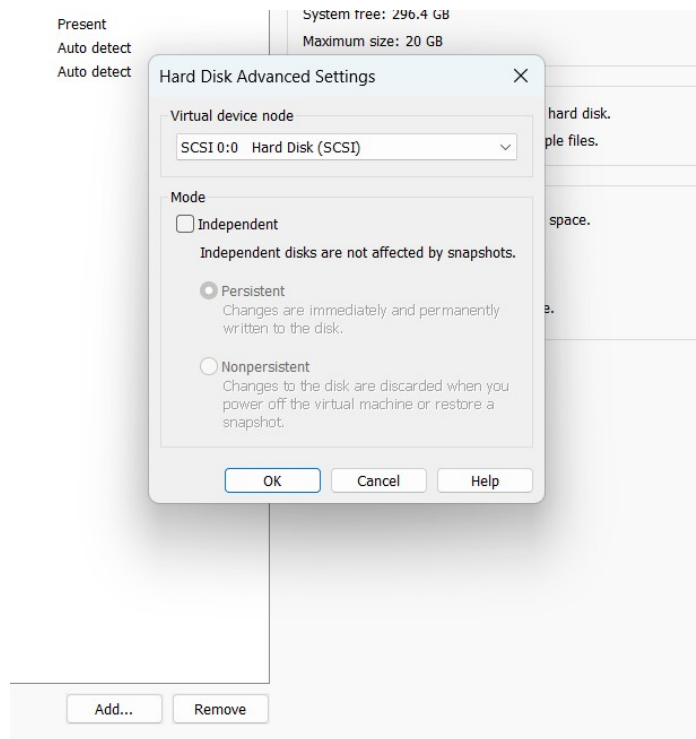
STEP 11: Verify that the newly created virtual hard disk is displayed in the Virtual Machine Settings.

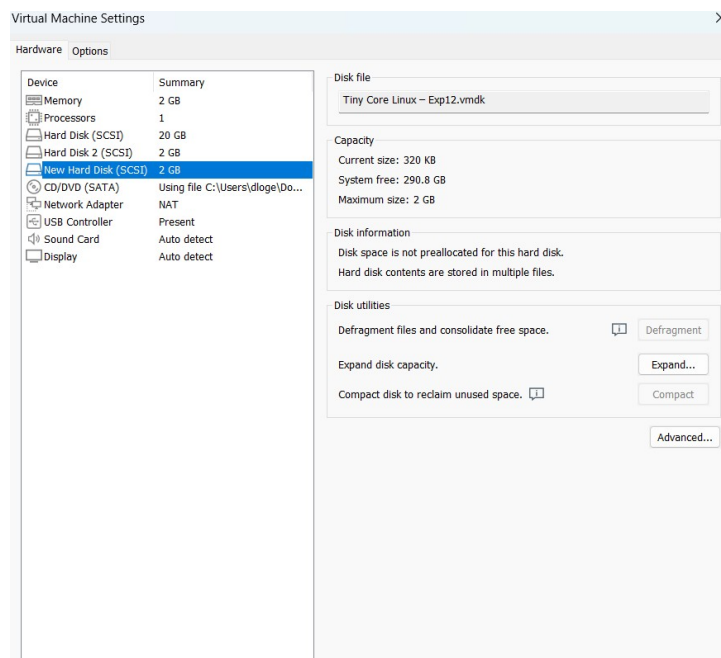
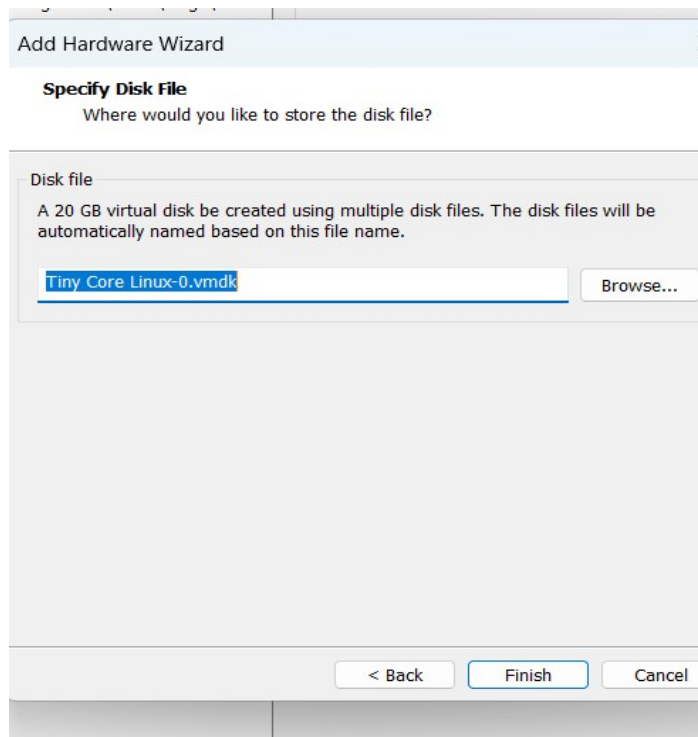
STEP 12: Click OK to save the configuration.

OUTPUT:









RESULT:

Thus, a 2 GB virtual hard disk was successfully created and allocated to the virtual machine using VMware Workstation.

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EXP NO: 10

CREATE A SNAPSHOT OF A VM AND TEST IT BY LOADING THE PREVIOUS VERSION/CLONED VM

AIM:

To create a snapshot of a virtual machine and test it by restoring and loading the previous version of the virtual machine.

PROCEDURE:

STEP 1: Open VMware Workstation and select the existing virtual machine.

STEP 2: Start the virtual machine and create a file or folder on the desktop.

STEP 3: In VMware Workstation, select VM → Snapshot → Take Snapshot.

STEP 4: Enter a suitable snapshot name, such as Snapshot 1, and click Take Snapshot.

STEP 5: After taking the snapshot, make a change to the virtual machine, such as modifying or deleting the file created earlier.

STEP 6: Open VM → Snapshot → Snapshot Manager.

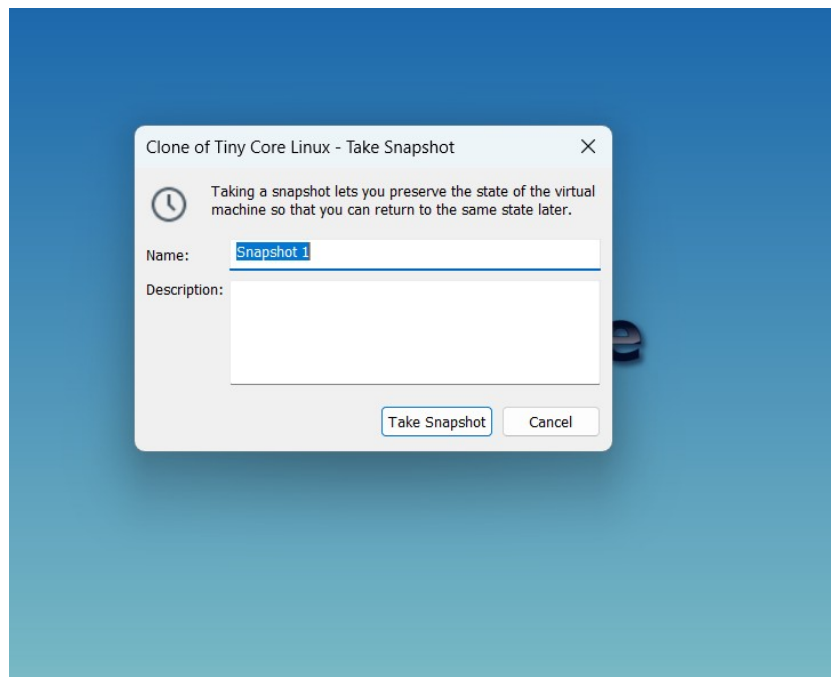
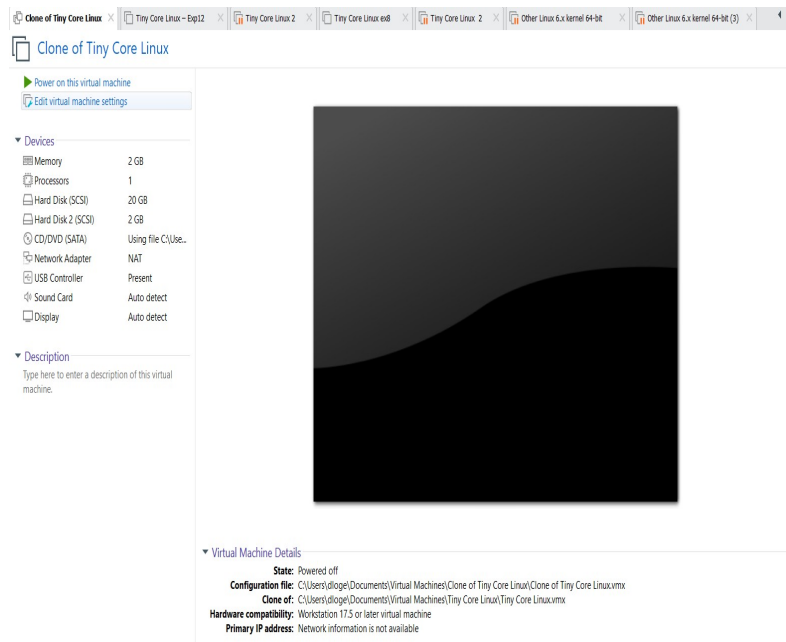
STEP 7: Select the previously created Snapshot 1 and choose Go To/Revert.

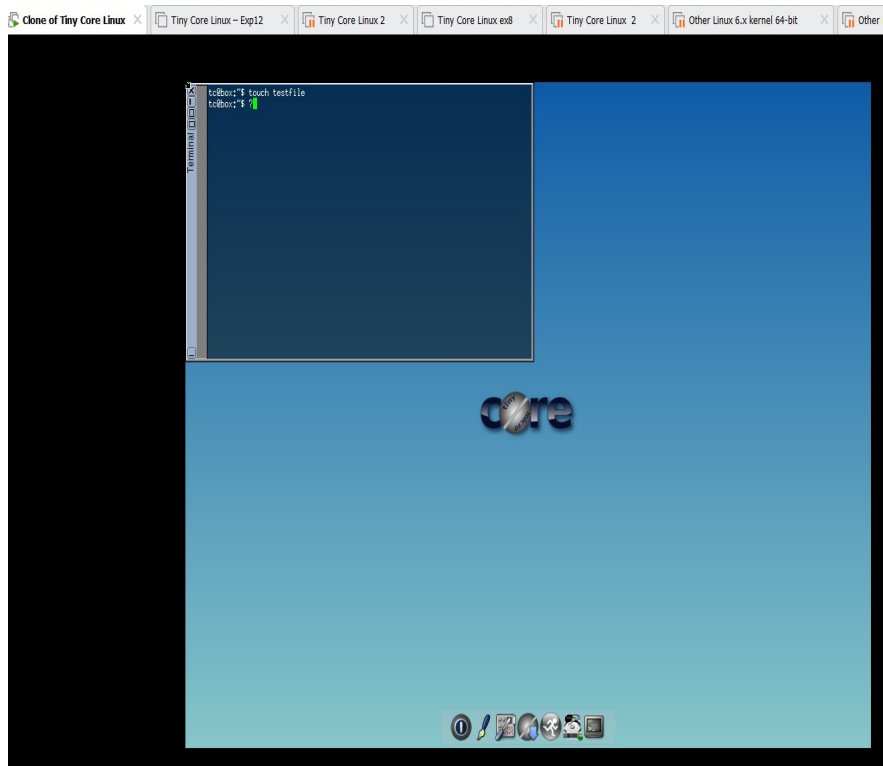
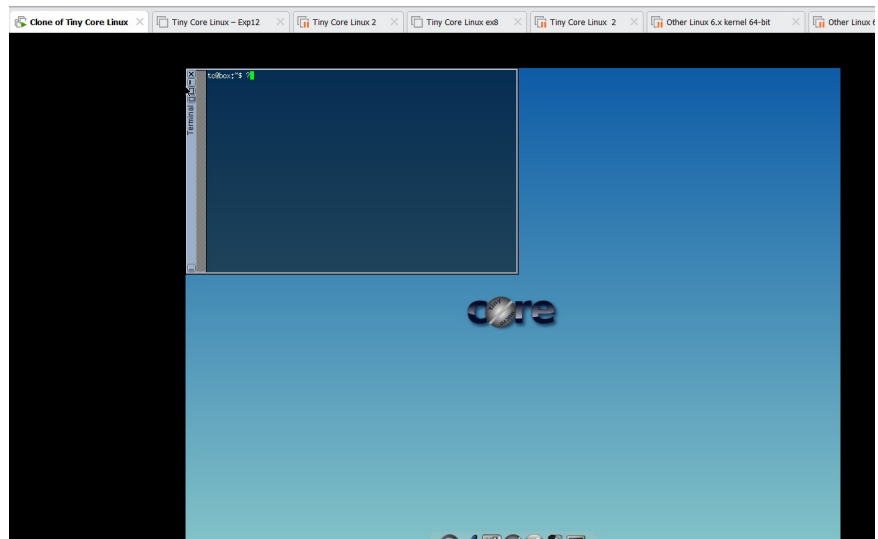
STEP 8: Confirm the restore operation and wait for the virtual machine to return to the snapshot state.

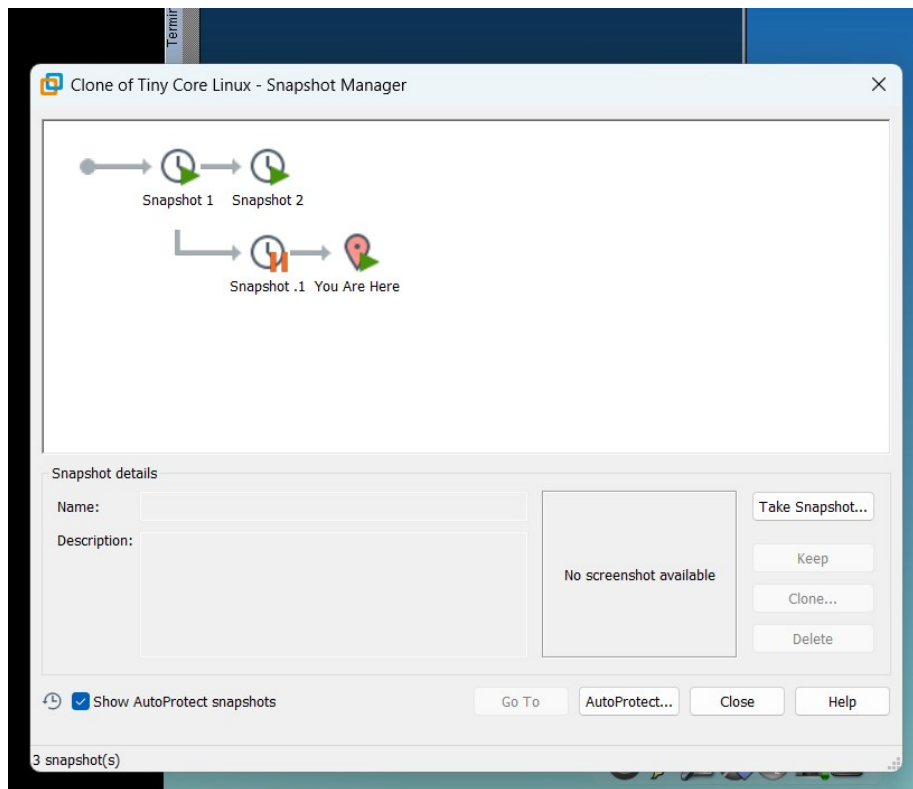
STEP 9: Check the desktop and verify that the file or previous configuration is restored.

STEP 10: Verify that the virtual machine has successfully returned to its earlier state.

OUTPUT:







RESULT:

Thus, a snapshot of the virtual machine was successfully created and tested by restoring the VM to its previous state.

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EXP NO: 11

CREATE A CLONING OF A VM AND TEST IT BY LOADING THE PREVIOUS VERSION/CLONED VM

AIM:

To create a clone of an existing virtual machine and test it by loading and running the cloned virtual machine using VMware Workstation.

PROCEDURE:

STEP 1: Open VMware Workstation and select the existing virtual machine.

STEP 2: Go to VM → Manage → Clone.

STEP 3: In the Clone Virtual Machine Wizard, click Next.

STEP 4: Select the current state of the virtual machine as the source and click Next.

STEP 5: Select Full Clone and click Next.

STEP 6: Enter a suitable name for the cloned virtual machine and select the storage location.

STEP 7: Click Finish to start the cloning process.

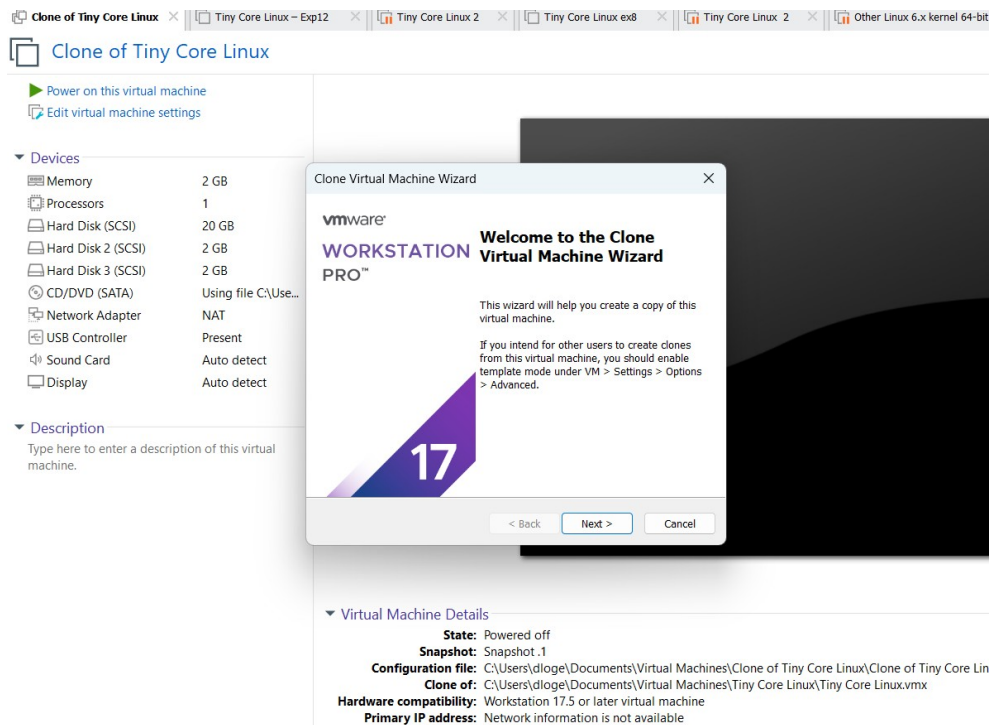
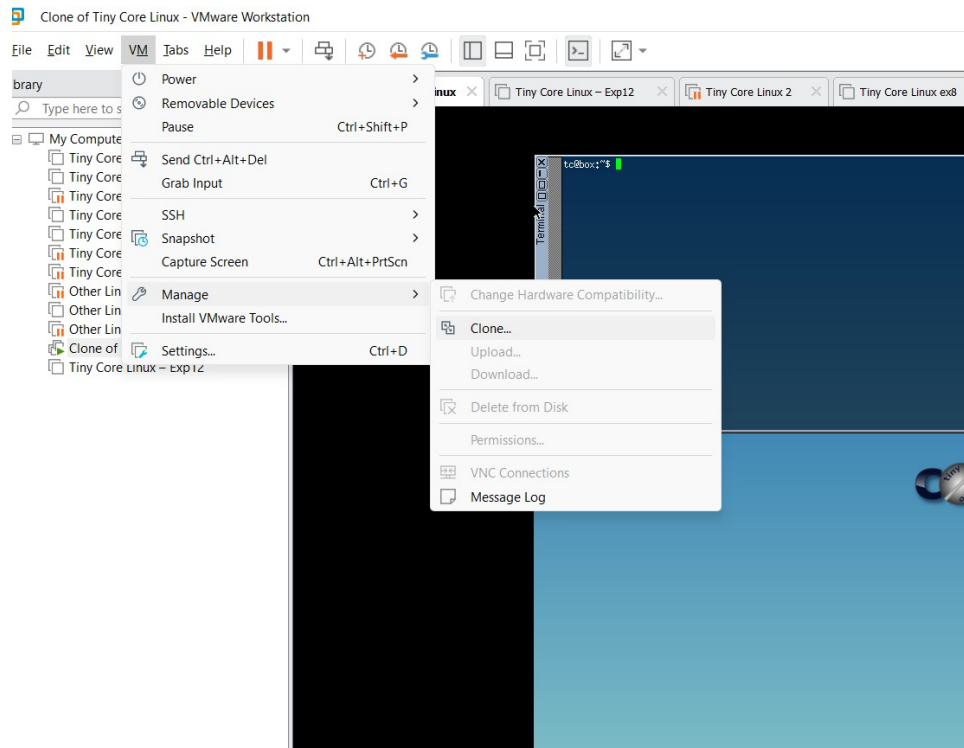
STEP 8: Wait until the cloning process is completed successfully.

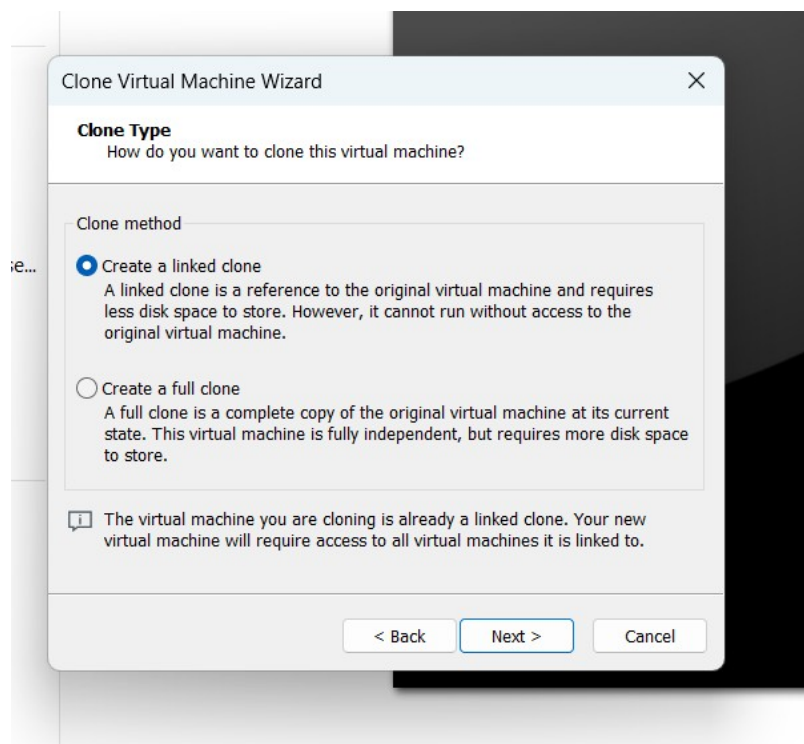
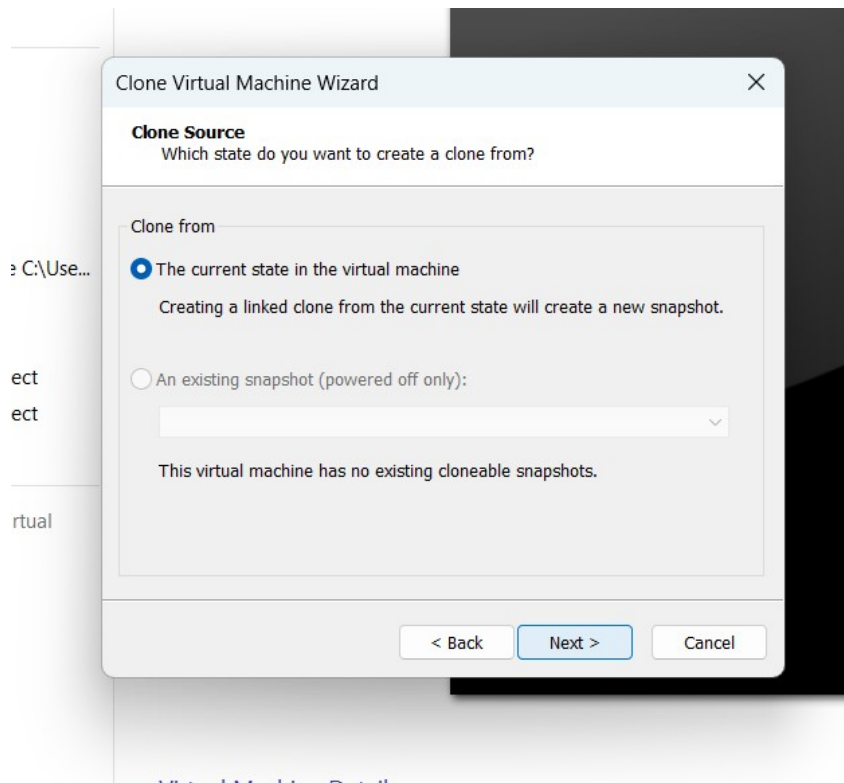
STEP 9: Open the newly created cloned virtual machine in VMware Workstation.

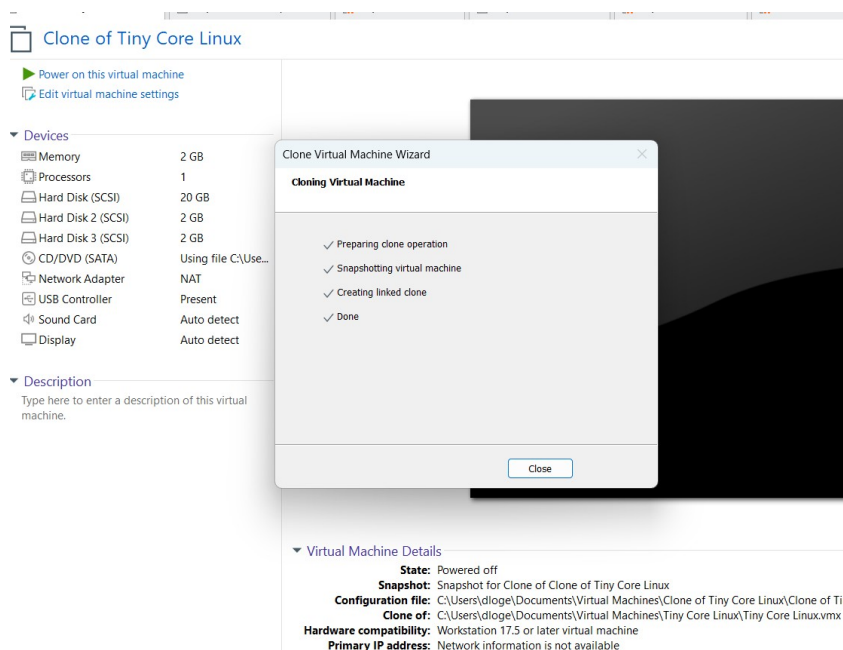
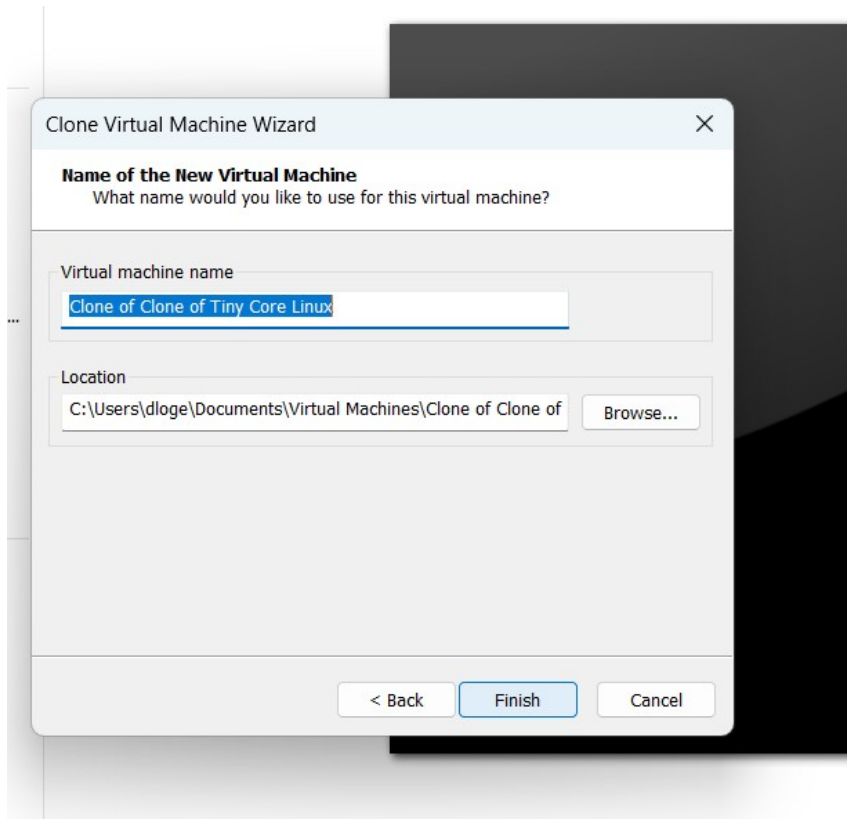
STEP 10: Power on the cloned VM and verify that the guest operating system loads successfully.

STEP 11: Test the cloned VM and verify that it works independently from the original virtual machine.

OUTPUT:







RESULT:

Thus, a full clone of the virtual machine was successfully created, loaded, and tested using VMware Workstation.

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EXP NO: 12

CHANGE HARDWARE COMPATIBILITY OF A VM USING VMWARE WORKSTATION

AIM:

To change the hardware configuration of an already created and configured virtual machine using VMware Workstation by adding a new virtual hardware component.

PROCEDURE:

STEP 1: Open VMware Workstation and select the existing virtual machine.

STEP 2: Right-click the virtual machine and select Settings.

STEP 3: In the Virtual Machine Settings window, click Add to open the Add Hardware Wizard.

STEP 4: Select Hard Disk and click Next.

STEP 5: Select SCSI as the virtual disk type and click Next.

STEP 6: Select Create a new virtual disk and click Next.

STEP 7: Set the virtual disk size to 2.0 GB.

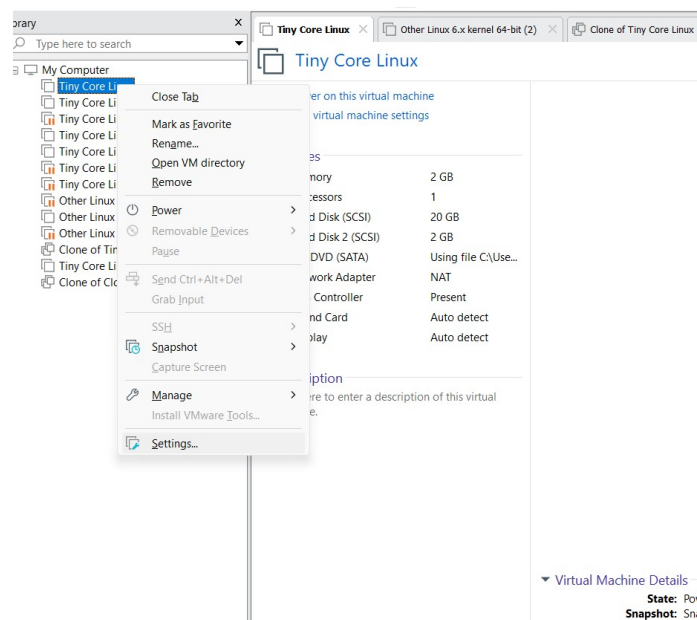
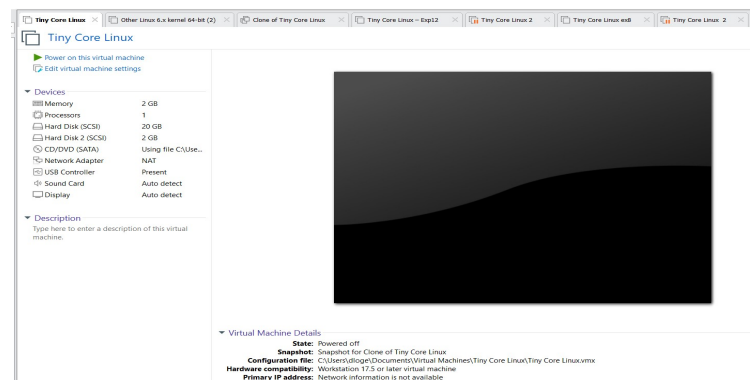
STEP 8: Select Split virtual disk into multiple files and click Next.

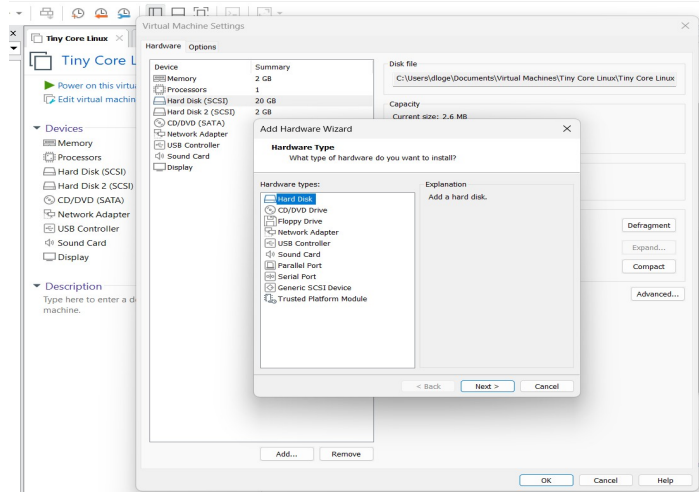
STEP 9: Give a suitable name to the virtual disk and click Finish.

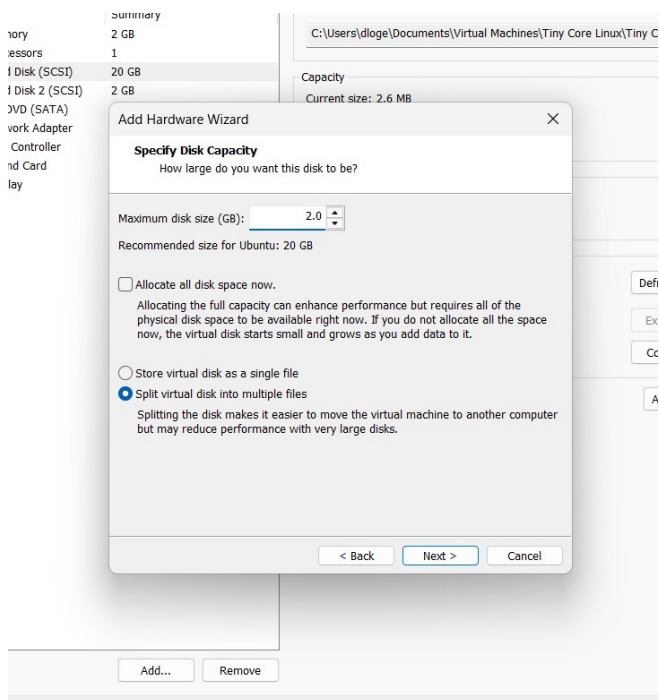
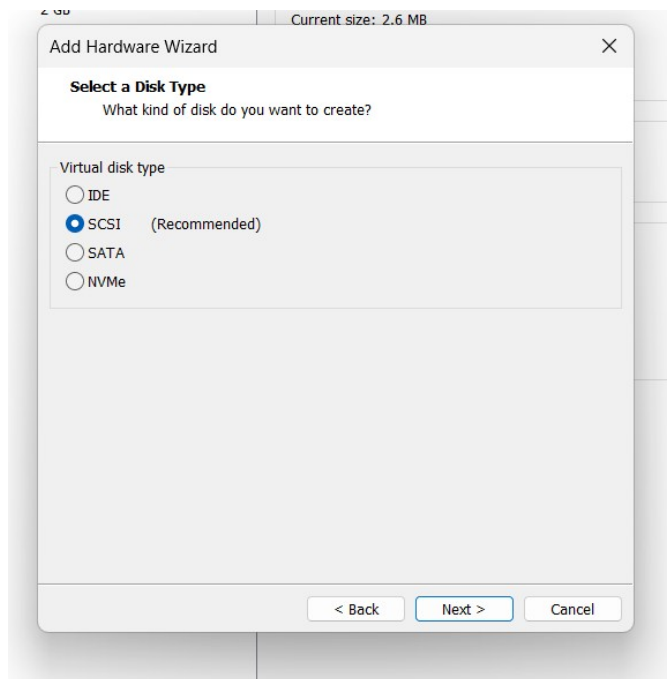
STEP 10: Verify that the newly added 2 GB SCSI virtual disk is displayed in the Virtual Machine Settings.

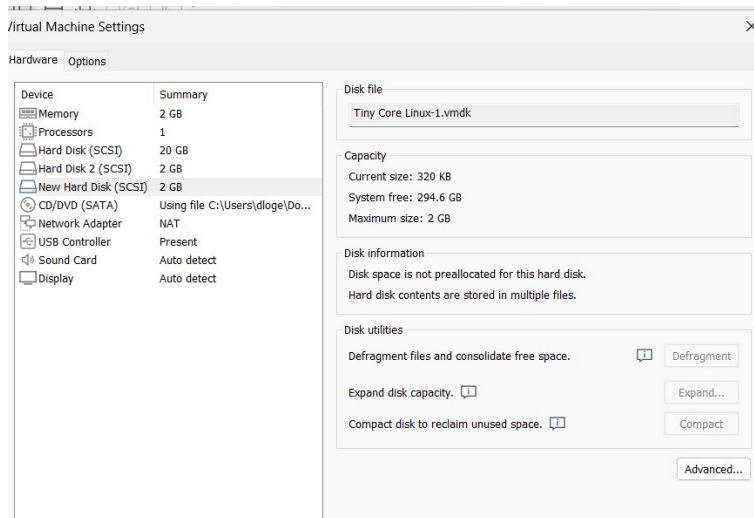
STEP 11: Click OK to save the changes.

OUTPUT:









RESULT:

Thus, the existing virtual machine was successfully configured with additional virtual hardware, and a 2 GB SCSI virtual disk was added using VMware Workstation.